LOG CANONICAL FANO PAIRS WITH ORBIFOLD FUNDAMENTAL GROUPS OF LARGE RANK

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Abstract. We show that a conjecture of Gachet, Liu, and Moraga on the orbifold fundamental groups of log canonical Fano pairs fails in any dimension at least 3: There exists a log canonical Fano variety whose orbifold fundamental group is free abelian of rank strictly larger than the dimension of the variety.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Gachet, Liu, and Moraga proposed the following conjecture on the orbifold fundamental groups of lc Fano pairs [GLM23].

Conjecture 1.1 ([GLM23, Conjecture 1])**.** *Let d be a positive integer. Then there exists a real number $c(d)$ depending only on d satisfying the following. For any lc Fano pair (X, B) of dimension d such that B is a $\mathbb{Q}$-divisor, the group $\pi_1^{\mathrm{orb}}(X, B)$ contains a normal abelian subgroup G such that*

$$\operatorname{rank} G \leq d \quad \text{and} \quad [\pi_1^{\mathrm{orb}}(X, B) : G] \leq c(d).$$

By [Bra21, Theorem 2], $\pi_1^{\mathrm{orb}}(X, B)$ is finite for any klt pair (X, B) such that X is projective, B is a $\mathbb{Q}$-divisor, and $-(K_X + B)$ is big and nef. For an integer $d \geq 2$, [FM23, Example 7.1] showed that the vertex of the affine cone over an abelian variety of dimension $d-1$ with respect to a very ample divisor is an lc singularity, and that the local fundamental group π_1^{loc} of the cone at the vertex is infinite and fits into an exact sequence

$$1 \to \mathbb{Z} \to \pi_1^{\mathrm{loc}} \to \mathbb{Z}^{2d-2} \to 1$$

such that the image of $\mathbb{Z}$ is contained in the center of π_1^{loc}.

(1) Let (X, B) be a dlt pair of dimension d such that X is projective, B is a $\mathbb{Q}$-divisor, and $-(K_X + B)$ is ample. If $\pi_1^{\mathrm{orb}}(X, B)$ is residually finite, that is, the normal subgroups of finite index in $\pi_1^{\mathrm{orb}}(X, B)$ have intersection $\{1\}$, then $\pi_1^{\mathrm{orb}}(X, B)$ contains a normal abelian subgroup G such that

$$[\pi_1^{\mathrm{orb}}(X, B) : G] \leq J(d),$$

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where $J(d)$ is a constant depending only on d [GLM23, Theorem G]. That theorem does not bound $\operatorname{rank} G$.

(2) For any lc Fano pair (X, B) of dimension 2 such that B is a $\mathbb{Q}$-divisor, $\pi_1^{\mathrm{orb}}(X, B)$ contains a normal subgroup G such that $[\pi_1^{\mathrm{orb}}(X, B) : G] \leq 7200$ and either G is abelian with $\operatorname{rank} G \leq 3$, or G is a quotient of the group

$$\langle x, y, z \mid [x, y] = [x, z] = 1,\ [y, z] = x^k \rangle$$

for some integer $k \geq 1$ [GLM23, Proposition 8.1].

Corti [Cor92, Example 16.4.4] showed that the projective cone X over a smooth projectively normal abelian surface in a projective space is normal and lc, and $\omega_X \cong \mathcal{O}_X(-1)$. Thus $(X, 0)$ is an lc Fano pair. That example does not consider $\pi_1^{\mathrm{orb}}(X, 0)$. In this paper, we show that Conjecture 1.1 does not hold for any integer $d \geq 3$ (Theorem 1.2). When $d = 3$, the variety in Theorem 1.2 is the cone of [Cor92, Example 16.4.4] over the abelian surface $C \times C \subset \mathbb{P}^8$, where $C \subset \mathbb{P}^2$ is a smooth cubic curve and $C \times C$ is embedded by the Segre embedding.

Theorem 1.2. *Let $d \geq 3$ be an integer, $C \subset \mathbb{P}^2$ a smooth cubic curve, and $N := 3^{d-1} - 1$. Let $A \subset \mathbb{P}^N$ be the image of C^{d-1} under the Segre embedding $(\mathbb{P}^2)^{d-1} \to \mathbb{P}^N$. Let $\mathbb{P}^N \subset \mathbb{P}^{N+1}$ be a hyperplane, $X \subset \mathbb{P}^{N+1}$ the projective cone over A with vertex $v \in \mathbb{P}^{N+1} \backslash \mathbb{P}^N$, and $H := X \cap \mathbb{P}^N$. Then:*

(1) X is a normal projective variety of dimension d, $\operatorname{Sing} X = \{v\}$, and $K_X \sim -H$. In particular, K_X is Cartier and $-K_X$ is ample.

(2) $(X, 0)$ and $(X, \frac{1}{3}H)$ are lc Fano pairs.

(3) Let X_{reg} be the non-singular locus of X. Then

$$\pi_1^{\mathrm{orb}}(X, 0) = \pi_1^{\mathrm{orb}}(X, \tfrac{1}{3}H) = \pi_1(X_{\mathrm{reg}}) \cong \mathbb{Z}^{2d-2}.$$

(4) $\operatorname{rank} G = 2d - 2 > d$ for any subgroup $G \subset \pi_1(X_{\mathrm{reg}})$ of finite index.

In particular, Conjecture 1.1 fails in dimension d.

Remark 1.3. The group $\pi_1(X_{\mathrm{reg}}) \cong \mathbb{Z}^{2d-2}$ in Theorem 1.2 is abelian, so it is a normal abelian subgroup of itself of index 1. Hence for the lc Fano pairs $(X, 0)$ and $(X, \frac{1}{3}H)$ in Theorem 1.2, only the rank bound in Conjecture 1.1 fails. By Proposition 3.2, $(X, 0)$ and $(X, \frac{1}{3}H)$ are not klt. Since the coefficients of their boundaries are < 1, they are not dlt either, and [GLM23, Theorem G] does not apply to them.

Sketch of the proof. The variety X is normal, $X \backslash \{v\}$ is the total space of the line bundle $\mathcal{O}_A(1)$ over A with zero section H, $K_X \sim -H$, and $(X, 0)$ and $(X, \frac{1}{3}H)$ are lc and not klt (see Proposition 3.2), which implies Theorem 1.2(1)(2). On the neighborhood $X \backslash H$ of v, both pairs restrict to $(X \backslash H, 0)$, which is lc and not klt since $X \backslash H$ is the affine cone over A with respect to $\mathcal{O}_A(1)$ and $K_A \sim 0$ [Kol13, Lemma 3.1].

Since $X_{\mathrm{reg}} = X \backslash \{v\}$ deformation retracts onto $H \cong A$, the standard approximations (see Definition 2.3) of 0 and $\frac{1}{3}H$ are 0, and the loop around H is trivial in $\pi_1(X_{\mathrm{reg}})$, we have

$$\pi_1^{\mathrm{orb}}(X, 0) = \pi_1^{\mathrm{orb}}(X, \tfrac{1}{3}H) = \pi_1(X_{\mathrm{reg}}) \cong \pi_1(A) \cong \mathbb{Z}^{2d-2}$$

(see Section 4). Any subgroup of finite index of $\mathbb{Z}^{2d-2}$ is isomorphic to $\mathbb{Z}^{2d-2}$, so its rank is $2d - 2 > d$.

The affine cone $X \backslash H$ is the cone of [FM23, Example 7.1] with vertex v, and $\pi_1(X_{\mathrm{reg}})$ is the quotient of the local fundamental group π_1^{loc} of $X \backslash H$ at v by the image of $\mathbb{Z}$.

Structure of the paper. In Section 2 we recall preliminaries, in particular orbifold fundamental groups and the rank of a group. In Section 3 we study the normality, the non-singular locus, the canonical class, and the singularities of the projective cone X. In Section 4 we prove the main theorem.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. A pair (X, B) is an *lc Fano pair* if X is projective, (X, B) is lc, and $-(K_X + B)$ is ample.

[GLM23, Conjecture 1] considers lc Fano pairs, which are not defined in [GLM23], and we take them in the usual sense of Definition 2.2.

2.2. Orbifold fundamental groups.

Definition 2.3 ([GLM23, Section 2])**.** Let X be a normal projective variety with non-singular locus X_{reg}, and $B = \sum_{i=1}^m b_i B_i$ a $\mathbb{Q}$-divisor on X, where $m \geq 0$ is an integer, $B_1, \ldots, B_m$ are distinct prime divisors, and $b_i \in (0, 1]$ for each i.

(1) For any real number $b \in [0, 1]$, the *standard round-down* of b is

$$b^{\mathrm{st}} := \max\left\{1 - \frac{1}{n} \middle| n \in \mathbb{N}^+ \cup \{+\infty\},\ 1 - \frac{1}{n} \leq b\right\},$$

where $\frac{1}{+\infty} := 0$. The *standard approximation* of B is $B^{\mathrm{st}} := \sum_{i=1}^m b_i^{\mathrm{st}} B_i$.

(2) For each i such that $b_i^{\mathrm{st}} < 1$, let n_i be the positive integer such that $b_i^{\mathrm{st}} = 1 - \frac{1}{n_i}$, and let $\gamma_i \in \pi_1(X_{\mathrm{reg}} \setminus \operatorname{Supp} B^{\mathrm{st}})$ be the class of a loop around B_i as in [GLM23, Appendix A]. The *orbifold fundamental group* of (X, B) is

$$\pi_1^{\mathrm{orb}}(X, B) := \pi_1(X_{\mathrm{reg}} \setminus \operatorname{Supp} B^{\mathrm{st}}) / \langle\langle \gamma_i^{n_i} \mid 1 \leq i \leq m,\ b_i^{\mathrm{st}} < 1 \rangle\rangle,$$

where $\langle\langle \cdot \rangle\rangle$ denotes the normal subgroup generated by the elements inside.

Definition 2.4. For any group G, the *rank* of G is

$$\operatorname{rank} G := \min\{\#S \mid S \subset G \text{ is finite and generates } G\} \in \mathbb{N} \cup \{+\infty\},$$

where $\min \emptyset := +\infty$.

3. Singularities and the canonical class of the cone

Construction 3.1. Let d, C, N, A, X, v, H be as follows.

(1) $d \geq 3$ is an integer.
(2) $C \subset \mathbb{P}^2$ is a smooth cubic curve.
(3) $N := 3^{d-1} - 1$.
(4) $A \subset \mathbb{P}^N$ is the image of C^{d-1} under the Segre embedding $(\mathbb{P}^2)^{d-1} \to \mathbb{P}^N$.
(5) $X \subset \mathbb{P}^{N+1}$ is the projective cone over A with vertex $v \in \mathbb{P}^{N+1} \backslash \mathbb{P}^N$, where $\mathbb{P}^N \subset \mathbb{P}^{N+1}$ is a hyperplane.
(6) $H := X \cap \mathbb{P}^N$.

Proposition 3.2. *Notation and conditions as in Construction 3.1. Then:*

(1) X is normal and $X \backslash \{v\} \cong \mathrm{Tot}(\mathcal{O}_A(1))$, where $\mathrm{Tot}(\mathcal{O}_A(1))$ is the total space of the line bundle $\mathcal{O}_A(1)$ over A.
(2) $K_X \sim -H$.
(3) $(X, 0)$ and $(X, \frac{1}{3}H)$ are lc and not klt, and $\mathrm{Sing}\, X = \{v\}$.

Proof. We choose homogeneous coordinates $[x_0 : \cdots : x_{N+1}]$ on $\mathbb{P}^{N+1}$ such that $\mathbb{P}^N = \{x_{N+1} = 0\}$ and $v = [0 : \cdots : 0 : 1]$. Let $X^\circ := X \backslash \{v\}$ and let

$$q : X^\circ \to A, \quad [x_0 : \cdots : x_{N+1}] \to [x_0 : \cdots : x_N]$$

be the projection.

(1) Since $C \subset \mathbb{P}^2$ is projectively normal, the Künneth formula implies that $A \subset \mathbb{P}^N$ is projectively normal, so

$$X \backslash H = X \cap \{x_{N+1} \neq 0\} \cong \mathrm{Spec} \bigoplus_{m \geq 0} H^0(A, \mathcal{O}_A(m))$$

is normal. In other words, $X \backslash H$ is the affine cone over A with conormal bundle $\mathcal{O}_A(1)$ in the sense of [Kol13, 3.8].

The projection q from v realizes X° as the total space $\mathrm{Tot}(\mathcal{O}_A(1))$ with zero section H, with fiber coordinate x_{N+1}/x_j over $A \cap \{x_j \neq 0\}$ for $0 \leq j \leq N$. Since A is smooth, X° is smooth. Since $X = (X \backslash H) \cup X^\circ$, X is normal. This implies (1).

(2) Since q is smooth of relative dimension 1, there exists an exact sequence

$$0 \to q^* \Omega_A \to \Omega_{X^\circ} \to \Omega_{X^\circ/A} \to 0.$$

For $0 \leq j, k \leq N$, on $\{x_j \neq 0\} \cap \{x_k \neq 0\}$ the local generators of $\Omega_{X^\circ/A}$ satisfy

$$d(x_{N+1}/x_k) = (x_j/x_k) d(x_{N+1}/x_j),$$

so $\Omega_{X^\circ/A} \cong q^* \mathcal{O}_A(-1)$. Since H is the zero locus of $x_{N+1}|_X \in H^0(X, \mathcal{O}_X(1))$ and q is the linear projection from v,

$$\mathcal{O}_{X^\circ}(H) \cong \mathcal{O}_{X^\circ}(1) \cong q^* \mathcal{O}_A(1).$$

Taking determinants, we get

$$K_{X^\circ} \sim q^* K_A - H|_{X^\circ}.$$

By adjunction, $K_C \sim (K_{\mathbb{P}^2} + C)|_C \sim 0$, so $K_A \sim 0$ and $K_{X^\circ} \sim -H|_{X^\circ}$. Since X is normal and $d \geq 3$, the restriction $\mathrm{Cl}(X) \to \mathrm{Cl}(X^\circ)$ is an isomorphism. Thus $K_X \sim -H$.

(3) Since H is a hyperplane section of X, $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$, so H is Cartier. By (2), K_X is Cartier, so $(X, 0)$ and $(X, \frac{1}{3}H)$ are pairs. Since A is smooth and $K_A \sim 0$, by (1) and [Kol13, Lemma 3.1(3)(5)], $(X \backslash H, 0)$ is lc and not klt.

Since X° is smooth, $(X^\circ, 0)$ is klt. Since $X = (X \backslash H) \cup X^\circ$, $(X, 0)$ is lc and not klt. Since $H \subset X^\circ$ is a smooth prime divisor, $(X^\circ, \frac{1}{3}H)$ is klt. Since $(X, \frac{1}{3}H)$ restricts to $(X \backslash H, 0)$ on $X \backslash H$, $(X, \frac{1}{3}H)$ is lc and not klt. Since $(X^\circ, 0)$ is klt and $(X, 0)$ is not klt, $(X, 0)$ is not klt at v. Since $(X, 0)$ is klt near any smooth point of X, $v \in \mathrm{Sing}\, X$. Since X° is smooth, $\mathrm{Sing}\, X = \{v\}$. □

4. Proof of the main theorem

Proof of Theorem 1.2. (1) Since the Segre embedding is a closed embedding, $A \cong C^{d-1}$ and $\dim A = d-1$. Since $X \subset \mathbb{P}^{N+1}$ is the projective cone over $A \subset \mathbb{P}^N$, X is a projective variety and

$$\dim X = \dim A + 1 = d.$$

By Proposition 3.2(1)(2), X is normal and $K_X \sim -H$. By Proposition 3.2(3), $\operatorname{Sing} X = \{v\}$. Since $H = X \cap \mathbb{P}^N$ is a hyperplane section of $X \subset \mathbb{P}^{N+1}$, $\mathcal{O}_X(H) \cong \mathcal{O}_X(1)$. Thus K_X is Cartier and $-K_X$ is ample.

(2) By Proposition 3.2(3), $(X,0)$ and $(X, \frac{1}{3}H)$ are lc. By (1), X is projective and $-K_X \sim H$ is ample. Since

$$-(K_X + \tfrac{1}{3}H) \sim_{\mathbb{Q}} \tfrac{2}{3}H,$$

$-(K_X + \frac{1}{3}H)$ is ample. Thus $(X,0)$ and $(X, \frac{1}{3}H)$ are lc Fano pairs.

(3) By (1) and Proposition 3.2(1), $X_{\mathrm{reg}} = X \backslash \{v\}$ is the total space of the line bundle $\mathcal{O}_A(1)$ over A, so $\pi_1(X_{\mathrm{reg}}) \cong \pi_1(A)$. Since C is a smooth projective curve of genus 1 and $A \cong C^{d-1}$,

$$\pi_1(X_{\mathrm{reg}}) \cong \pi_1(A) \cong \mathbb{Z}^{2d-2}.$$

For the boundary 0, we have $m = 0$, so $\pi_1^{\mathrm{orb}}(X,0) = \pi_1(X_{\mathrm{reg}})$ by Definition 2.3(2).

For the boundary $\frac{1}{3}H$, since H is a prime divisor on X, Definition 2.3 is used with $m = 1$, $B_1 = H$, and $b_1 = \frac{1}{3}$. Since

$$1 - \frac{1}{n} \geq \frac{1}{2} > \frac{1}{3} \quad \text{for any } n \in (\mathbb{N}^+ \backslash \{1\}) \cup \{+\infty\},$$

$b_1^{\mathrm{st}} = 1 - \frac{1}{1} = 0$ and $n_1 = 1$. Thus $(\frac{1}{3}H)^{\mathrm{st}} = 0$ and

$$\pi_1^{\mathrm{orb}}(X, \tfrac{1}{3}H) = \pi_1(X_{\mathrm{reg}}) / \langle\langle \gamma_1 \rangle\rangle,$$

where γ_1 is a loop around H in X_{reg}. Since $v \notin \mathbb{P}^N$, $v \notin H$, so $H \subset X_{\mathrm{reg}}$ and γ_1 is the boundary of a small disc in X_{reg} that is transversal to H at one point. Thus γ_1 is the trivial element of $\pi_1(X_{\mathrm{reg}})$, and $\pi_1^{\mathrm{orb}}(X, \frac{1}{3}H) = \pi_1(X_{\mathrm{reg}})$.

(4) Let $G \subset \pi_1(X_{\mathrm{reg}})$ be a subgroup of finite index. Since $\pi_1(X_{\mathrm{reg}}) \cong \mathbb{Z}^{2d-2}$, $G \cong \mathbb{Z}^{2d-2}$, so $\operatorname{rank} G = 2d-2 > d$ as $d \geq 3$.

Finally, by (1) and (2), $(X,0)$ is an lc Fano pair of dimension d, and 0 is a $\mathbb{Q}$-divisor. By (3) and (4), for any real number $c(d)$, there does not exist a subgroup $G \subset \pi_1^{\mathrm{orb}}(X,0)$ such that $\operatorname{rank} G \leq d$ and $[\pi_1^{\mathrm{orb}}(X,0) : G] \leq c(d)$. Therefore Conjecture 1.1 fails in dimension d. □

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